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A decision support tool for energy efficient synchromodal supply chains

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ABSTRACT

The industrial supply chain is a significant consumer of limited energy resources worldwide. Companies have been encouraged to reduce energy costs by restructuring their logistics network, improving their transportation asset fuel efficiency, and reducing travel distance. The recent evolution of synchromodal supply chains has focused on flexible transportation modes as a new option for supply chain managers to reduce the total energy cost. Illustrating the monetary advantages of such reconfigurations will guide decision makers to get the most from all available options.

In this paper, the process of modeling and developing a user-friendly, customizable tool for supply chain decision makers called the Synchromodal Supply Chain Energy Analysis tool (SySCEA) is presented. SySCEA enables the analysis of the overall environmental and economic effects due to supply chain fuel consumption and changing network configuration as it relates to order quantity, demand quantity, product size, packaging, mode and transportation assets. Currently, SySCEA supports up to five arc/routes, plus the first and last mile, for a synchromodal supply chain. However, due to the modular nature of the tool it is simple to extend it to larger networks.

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1. Introduction

Supply chains are a primary user of energy worldwide. Environmental managers need to consider all supply chain issues from materials to energy in order to maximize system performance (Clift and Wright, 2000). Historically supply chain design has focused on minimizing costs or maximizing delivery performance. However, emphasis on environmental factors when designing the network has increased. Yet what environmental factors to consider and quantify is a question that needs to be answered. Full Life Cycle Assessment (LCA) of products that focuses on the measurement of manufacturing and distribution processes is one of the most common approaches to addressing and quantifying environmental factors. The LCA approach is computationally intensive, especially when the whole supply chain is considered. Another approach is to consider a partial LCA and model select processes from the product life cycle, e.g., one can solely focus on the calculation of Green House Gases (GHG) resulting from the transportation of products in a supply chain. A second approach is to calculate the carbon footprint, which is the most popular metric to measure environmental impact caused by a supply chain. Fuel consumption is a good predictor for estimating carbon footprint (Eskandarpour et al., 2015).

Decreasing the fuel consumption caused by the logistics and transportation of products in a supply chain is not only beneficial monetarily, but it also reduces emissions and the reliance on fossil fuel. In fact, companies should be encouraged to consider rebuilding their logistic networks in order to improve the fuel efficiency of their transportation assets. The shipping of goods involves multiple corporate entities and the complexity of the overall supply chain requires suppliers and transport providers to look for ways to decrease costs, maximize energy efficiency, and plan for a more sustainable, greener future.

Multimodal transportation has great potential for improving the environmental performance of freight transportation, but typically supply chains rely on a single mode and the decision maker does not have the option of a mode choice (Fig. 1). This is primarily due to the lack of an integrated representation of how modal shifts affect supply chain performance. Synchromodality provides a framework for managing flexible modal shifts in supply chains. This concept makes the mode choice decision process more complex, but more flexible (Dong et al., 2017).

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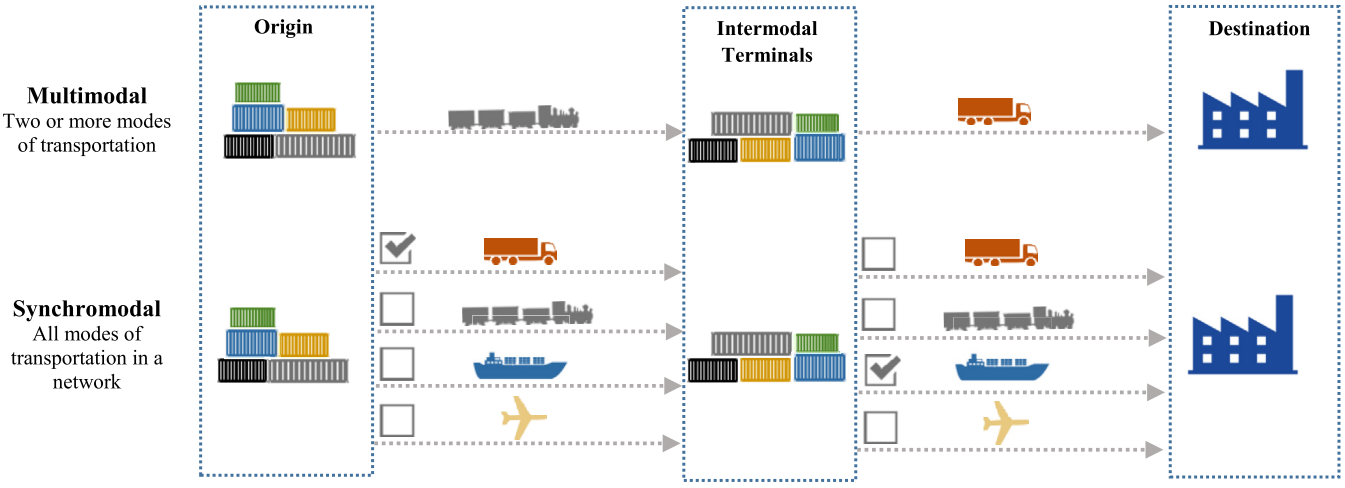


Fig. 1. Comparison of multimodal and synchromodal transportation regarding mode choice. (Modified from Verweij 2014).

To create a green, sustainable network, decision makers must balance the economic and environmental trade-offs in logistics (Harris et al., 2011). The synchromodal approach enables supply chains to have a flexible transport network and freely combine the best modes for each route. Using this approach, decision makers can optimize their transport process and reduce cost, while also taking control of carbon emissions by shifting to cleaner mode choices (Putz et al., 2015). In synchromodal supply chains, the goal is to find the best transport alternative in a multimodal network by addressing the objectives of cost reduction, on-time delivery, and emission reduction, as well as improving the robustness and accessibility of a network. Thus, synchromodality, along with other transportation factors, presents potential areas for energy saving.

Current energy consumption data shows that transportation consumes a large portion of overall energy usage in a supply chain. Fig. 2 depicts fuel consumption for all forms of transportation (e.g., roads, air, marine, and rail) and shows that it contributed 26% of total greenhouse gas emissions and was the second largest source of carbon dioxide (CO_2) emissions (31%). From 1990 to 2014, transportation emissions increased by 9% as a result of a range of short-term and long-term variables like economic growth, energy price fluctuation, and increased miles traveled by motor vehicles to meet people and customer needs (US Environmental Protection Agency, 2016).

Worldwide transportation of goods requires approximately 15 million barrels of oil a day, which is one-fifth of total world oil production (Meyer, 2009). Petroleum consumption in the form of gasoline and diesel is the main source of energy for transportation (Fig. 2). The switch to a more environmentally-friendly network that decreases the fuel consumption from transportation logistics is one way to reduce carbon emissions in the supply chain. However, to reduce energy usage and carbon emissions, the key is to develop metrics and evaluate the effectiveness of different alternatives (Cholette and Venkat, 2009). Although concerns about environmental preservation have increased, the lack of adequate managerial models is evident (Ferretti et al., 2007). Fig. 3 illustrates how changing logistics factors can lead to a more fuel efficient supply chain. The overall potential for fuel saving by changing the supply chain logistic strategy is more than 23% over the 2008 baseline.

Supply chain managers focus on cost reduction in order to stay competitive in the market. However, following an environmentally friendly strategy can be costly in the short run. In recent years, studies have focused on making supply chains greener by finding the balance between economic and environmental efficiency (Ferretti et al., 2007).

In this paper, we explore the impact of reducing energy consumption/cost in synchromodal supply chain design and operation. The model developed uniquely considers both energy efficiency issues and supply chain design parameters to obtain new insights to support decision making under changing energy cost scenarios. A user-friendly Excel tool was developed to assist supply chain managers in making decisions. The Synchromodal Supply Chain Energy Analysis (SySCEA) tool allows analysis of a five-route/six-node based supply chain, with both first and last mile transportation, for four interchangeable modes of transportation (road, rail, air and water). SySCEA considers order quantity, different packaging and container sizes to estimate the annual energy cost of the supply chain. SySCEA is customizable and provides energy consumption data, energy costs and carbon emission results for the supply chain.

2. Literature review

Energy management is a broad area and has several different aspects especially when the whole supply chain is considered. Although decision-makers know that strategic decisions are more effective than operative decisions environmentally, it is still hard to

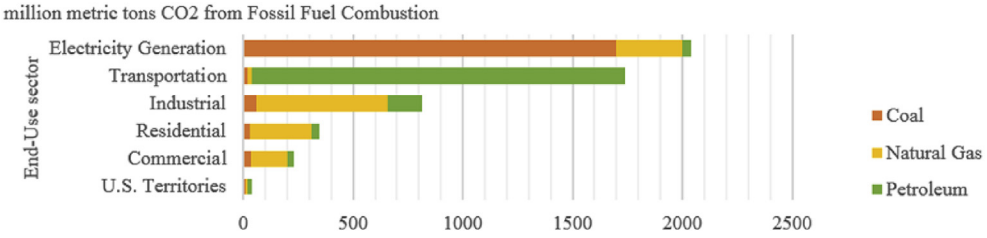


Fig. 2. 2014 The end-use sector's CO2 emissions from fossil fuel combustion in the US (US Environmental Protection Agency, 2016).

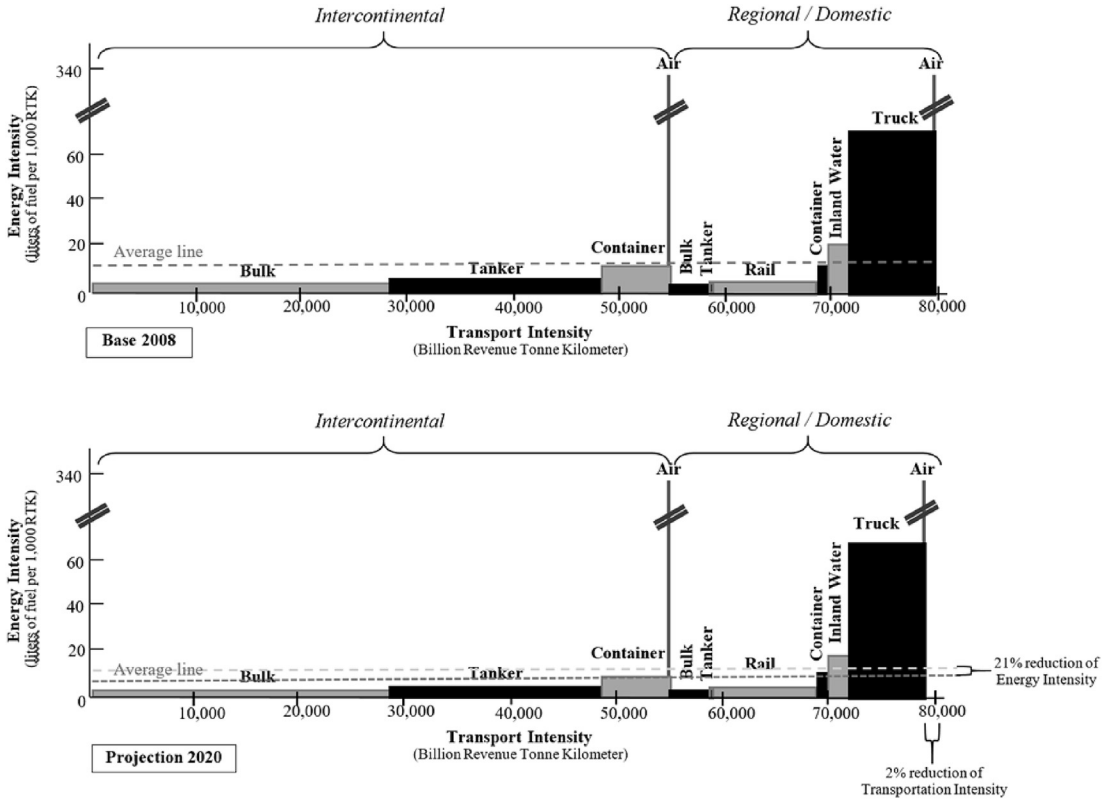


Fig. 3. Potential energy savings in a supply chain (Meyer, 2009).

say what decision has the largest impact and to what extent (Harris et al., 2011). In such a complex environment, taking advantage of decision support systems can help supply chain managers improve the performance of their network design. Decision making for synchromodal supply chains are even more challenging because they include many factors that can be changed simultaneously.

In this literature review, the focus is on energy saving models for supply chains. Subsection 2.1 covers energy savings models within a supply chain with a special focus on logistics and transportation. In section 2.2, methods for calculation of energy consumption, with a special focus on energy saving in multimodal and synchromodal transportation systems are discussed. Also, available decision support tools developed to measure and model environmental impact of transportation systems are explored. Finally, Section 2.3 brings attention to the literature gaps previously overlooked and how they are addressed in this paper.

2.1. Models of energy savings within a supply chain

Supply chain environmental factor measurements have mostly been conducted using Life Cycle Assessment (LCA), but LCA approaches tend to be expensive, time-consuming and need expertise in environmental management. The most common methods of LCIA are Eco-Indicator 99 and Impact 2002+. The most recent method is the ReCipe that provides a single score. However, instead of adopting these methods, another approach would be to conduct assessments that focus on the partial assessment of environmental factors choosing a single scope such as facilities, transport, process or product design (Eskandarpour et al., 2015).

Among all introduced environmental factors, transportation in a supply chain, specifically in vast countries like the USA, can be very energy-intensive. Bouzembrak et al. (2013) recommend focusing more on minimizing transportation cost above other costs when reducing the energy cost of the supply chain. Heller and Keoleian (2003) in their study estimated the use of diesel trucks and assign 25% of the total energy consumption to the operation of diesel trucks. Finally, reducing the consumption of transportation fuel is one of the quickest paths to making supply chain operations greener.

Focusing on multimodal freight movement also can save both fuel and energy. According to Kapetanis et al. (2016), the recent evolution of moving from multimodality to synchromodality in a supply chain brings the flexibility of considering the most efficient mode of transportation when configuring a network. Zhang and Pei (2016) believe that what is missing in the literature is a strategic planning model that considers the operational level and gives the user capability to compare between different modes. Also, Dong et al. (2017) mention that synchromodality research is still in an early stage and the first quantitative studies of this framework began in 2015. Since 2015 the interest in its implementation is growing within the transportation and logistics areas. The primary benefit of this concept is improving sustainably in a supply chain both economically and environmentally which come from a modal shift in real-time transportation planning. Unfortunately, the relationship between synchromodality and supply chain management processes has not been addressed in the literature, so innovative approaches to address this gap are needed.

Reviewing the models in this field show that quantitative energy-saving models that focus on transportation and logistics have grown in the last two last decades. In a review of green and sustainable supply chains from 1990 to 2013, Seuring (2013) found 308 papers of which just 36 of them built or used quantitative models (approximately 12%). Among these 36 papers, 12 papers discussed energy-related subjects and four papers focused specifically on transportation energy saving (Khoo et al., 2001; Sonesson and Berlin, 2003; Ferretti et al., 2007; Cholette and Venkat, 2009). Khoo et al.

(2001) encouraged work on models that would change customer orders and demands for calculations of supply chain energy costs. They recognized a need for intensive research in this field. They also provided the option of more fuel-efficient modes of transportation considering its associated cost. To estimate the energy consumption of a supply chain, Sonesson and Berlin (2003) focused on the calculation of total energy used in dairy products and showed how important the packaging materials and transportation emissions of NOx are in total energy use. The paper focused on generated scenarios by a reference group and the model cannot be used for general supply chains. Ferretti et al. (2007) created a mathematical model to analyze the economic and environmental impact in the aluminum supply chain, more specifically in two chains in Lombardy, Italy. The paper is a case study on the aluminum supply chain that focuses on saving raw materials and electrical energy in the transportation of aluminum alloy by suggesting the transfer of molten aluminum rather than solid ingot. This paper combined costs of metal companies, transportation cost, and the industry's pollution costs, but did not analyze them separately. It considers one mode of transportation over a limited distance. This paper points out that the shape of the product is relevant to transportation efficiency; e.g., in ladle-based transport, less material can be moved in comparison with ingots. Cholette and Venkat (2009) concluded that most of the energy-related supply chain research studies have provide insights on environmental issues and energy efficiency in the supply chain logistics, but adoption in practice has been limited.

Clift and Wright (2000) estimated two environmental impacts: energy and solid waste generation. They explored different scenarios to observe the savings in energy use. In another paper, Clift (2003) combined environmental indicators and economic performance for sustainable development of a supply chain. Later, Fichtner et al. (2004) developed an inter-firm energy supply methodology to show how a combined cycle power plant can meet specific energy demands and become more sustainable. Frota Neto et al. (2008) advocated changing the objective in the design of logistic networks from cost minimization to include both cost and environmental impact minimization. To do so, they created a framework for profitable, sustainable logistic networks considering environmental performance and cost efficiency. Cruz (2008) presented a model of multi-criteria decision-making behavior with the objective emission/waste and risk as part of the minimization and profit maximization for different decision makers in a supply chain.

Due to the complexity of energy modeling in a supply chain, finding an optimum solution is difficult, and decision-makers need to be engaged in choosing the option that fits their conditions. Cholette and Venkat (2009) considered the configuration of the supply chain structure and energy efficient inventory policies. The primary focus of this research was to show that different configurations in the supply chain can bring a wide range of results in energy and emission statistics. The paper did not consider effective use of product and packaging weight and volume or energy efficient transportation assets.

Edwards et al. (2009) focused on the “last mile” analyzing non-food shopping trips and deliveries in the UK based on carbon emissions. In this case, high energy intensity for online retailers is discussed. This study did not consider the whole supply chain and focuses on a small part of transportation.

Harris et al. (2011) developed a model that calculates the total cost of transportation and inventory. Their model considers the amount of transported goods—not life cycle energy consumption. The CO2 emission data covers a year for road transportation. Their focus was on one case study and the result was based on the defined scenario. The study attempted provide a decision support tool for strategic modeling, however, it cannot be used for general supply chains.

Bono et al. (2014) considered lead-time, distance, and weight in transportation energy cost. Emergency orders would lead to shorter lead-times but require more fuel consumption and costs. In this paper, the optimal inventory that minimizes the production and energy cost with emergency demands is discussed. This model considers air and ground transportation. The results show demand and product weight is the most significant factor in logistic costs.

The literature review of energy saving models reveals a quantitative modeling gap in synchromodal supply chains. Also, there is a lack of studies on the relationship of energy consumption, supply chain transportation, and inventory management. In models that have been developed for total energy costs in a supply chain, some decisions such as effective use of product and packaging weight and volume or energy efficient transportation assets are not considered.

In the following section, the details of previous studies that focus on providing energy saving calculations for supply chain decision makers are presented.

2.2. Energy calculation and decision support tools for supply chains

For partial LCA based measurement of environmental factors focusing on transportation, tools have been developed by national or regional organizations (Eskandarpour et al., 2015). Because of the variety of factors in environmental effect measurement, these tools are case-base and cannot be generalized to other supply chains. For example, Khoo et al. (2001) focused on a case study of an aluminum supply chain using simulation, which they referred to as the

Table 1
List of some freight energy consumption tools.

Tool Name	Characteristics					
	Modes				Result	Description
	Road	Rail	Air	Water		
Mobile 6.2 (Tang et al., 2004)	●				Calculates emissions factors in grams per mile for one heavy duty vehicle.	Uses average fuel economy for the entire national fleet for each vehicle category and model year
Clean Marine Energy Fuel and Emission Calculator (Clean Marine Energy, 2017)				●	Calculates fuel cost and emission for different types of vessels	The fuel and vessel type are the only variables calculates annual fuel price and emissions
Freight Emissions Calculator (EA Logistics, 2017)	●				Calculates CO2 emission	For Less than Truck Load and Truck load option considering given distance and weight
Carbon Emission Estimator (Union Pacific Railroad)		●			Calculates CO2 emission	Considers distance, commodity type and number of truckload
Emission Calculator (Multimodal Company, 2017)	●		●	●	Calculates CO2 emission	Just considers a full load standard container. In water way, it considers sea vessels

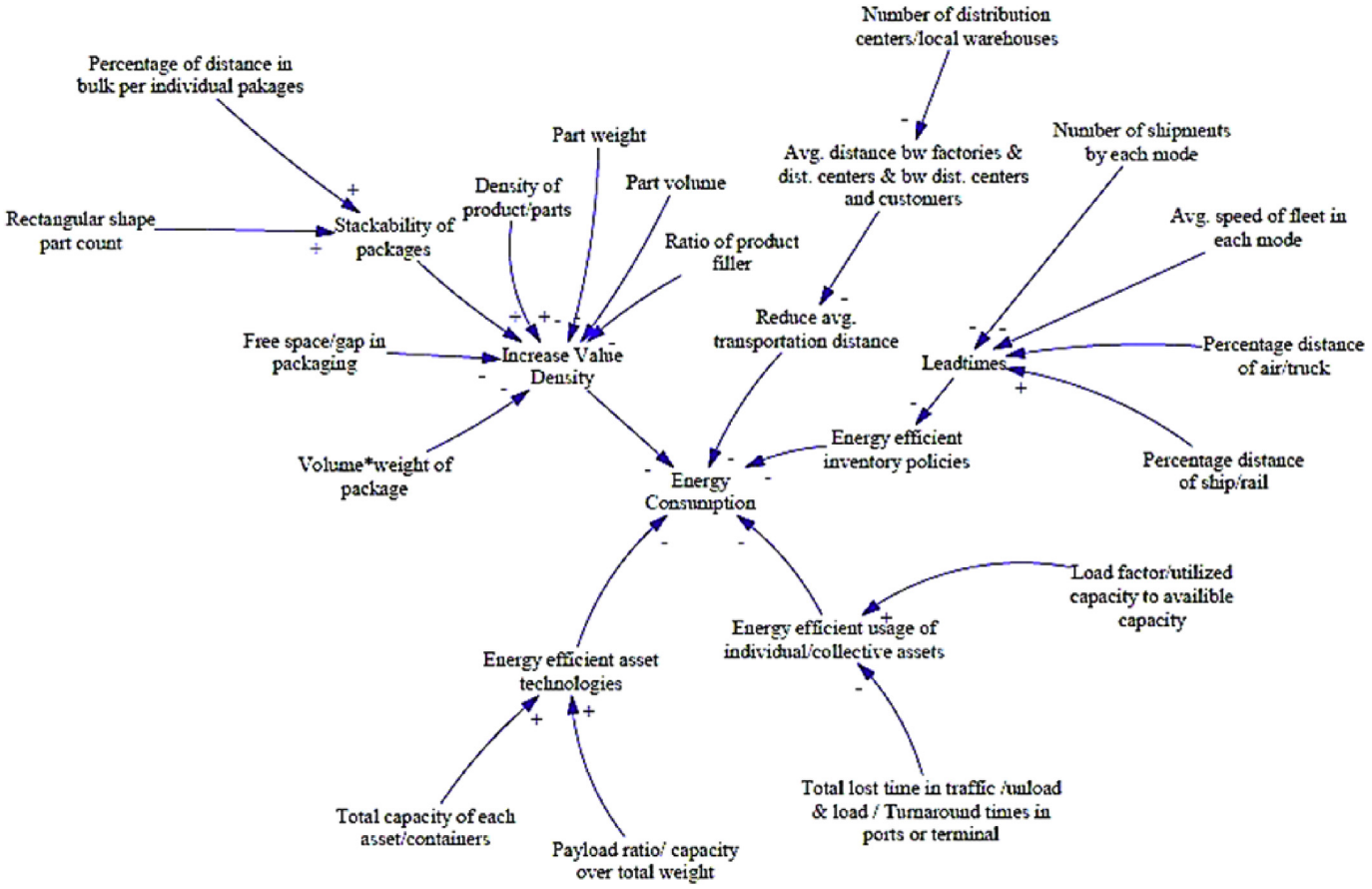


Fig. 4. The cause and effect diagram of energy saving in supply chain logistics.

ProcessModel 2000, to create a green supply chain. Sonesson and Berlin (2003) focused on analyzing the impact of different supply chain configurations in the dairy products supply chain using real data. To simulate different scenarios, they used an MFA/SFA model known as SAFT and the results were analyzed with LCA methodology. Cholette and Venkat (2009) also simulated cases in the USA wine industry where they analyzed distribution using a web-based tool called CargoScope. Dong et al. (2017) presented a case study to show how by increasing the modal share of rail and road a shipper can save logistics cost and reduce emissions. The focus of their study is on the inventory management of a firm and considers warehouse storage and handling cost, and the cost of capital for inventory. Their model uses simulation to reach a solution for modal split. Their model focuses on rail and road and doesn't discuss freight characteristics such as weight, size and packaging, and the variety of the assets. Zhang and Pel (2016) developed a strategic model of synchromodal freight transportation and tested it using the data from the Port of Rotterdam. Their model, named SynchroMO, considers demand and supply models for three modes: rail, road and in-land waterway, to calculate the total cost of shipment and total required time, network flow and emissions. Their study relaxes equipment and shipment size constraints by considering a full load standard container as the freight and doesn't consider other sizes of containers or less than truck load freights. Their model reveals some comparative results on the differences between an intermodality and synchromodality approach toward transportation for selected routes. Kapetanis et al. (2016) designed a real-time model that estimates time cost and emissions for transportation events for three modes: rail, road, and sea, for a given path (Shanghai–Piraeus–Prague). The tool is developed based on a case study and doesn't calculate energy consumption cost. The tool also relies on historical data of the selected ports.

Based on our study on estimating energy cost techniques, simulation is the most common way to handle this problem in the literature. Although simulation is a good way to estimate energy cost, it is time-consuming, expensive and highly scenario dependent. Moreover, simulation needs professional software and modeler expertise to be done correctly. Thus, this method cannot be used by every decision-maker associated with the supply chain.

Most of the tools developed as carbon emission calculators have focused on urban transportation rather than supply chain logistics. In other words, they are mainly passenger-based rather than freight based.[1] There are a few freight-based tools available for public use, but these tools have focused on just one mode of transportation and not the whole supply chain.

The tools in Table 1 do not give the option for the user to customize the freight characteristics, nor to understand the rationale behind the assumptions. Many of these tools have assumptions that lead to unrealistic results. For example, in Clean Marine Energy Fuel and Emission Calculator (Clean Marine Energy, 2017) the container size is not considered or in the Emission Calculator (Multimodal Company, 2017) only a standard full load container is considered and it doesn't evaluate other options.

Overall, the review of energy calculation tools and models for supply chains shows that these models and tools are mainly scenario-base and cannot be applied to a general supply chain.

[1] Examples: https://www3.epa.gov/carbon-footprint-calculator/and https://transportation-forms.stanford.edu/cost/.

Table 2
Variables of energy consumption in supply chain.

0	1	2	3	Decision Factors
Supply Chain Set Up	Increased Value Density	Product	Filler	Density/Ratio Ratio of Filler
			Assembly	Parts' Volume Parts' Weight Density of Parts
		Packaging	Layers	Stackability Free Space/Gap in Pallet
			Shape	Rectangular Shape Part Count
			Volume Weight	Throwaway/Damping Material Volume*Weight
	Reduced Average Transportation Distance	Assembly Postponement/ Nearshoring of Sourcing and Production	Delayed Pack	Percentage/Difference of Distance in Bulk per Individual Packaging
		Distribution Network Structure	_	# Distribution Centers/Local Warehouses Avg. Distance Between Factories and Distribution Centers Avg. Distance Between Distribution Centers and Customers
	Changing Mix of Transportation Modes	Outright Shift	Inventory Modes of Transportation	Lead Times # Shipments by Air Freight/Truck per Unit Time Av. Speed of Fleet in Each Mode
		Changing sequence	_	Percentage Distance of Each Mode of Transportation
Transport assets	Address Asset Technology	Scale	_	Total Capacity of Each Asset/ Containers
		Relative Drag/Propulsion Systems	Technology Energy Usage	Efficiency of Energy Usage
		Payload Ratio	_	Ratio of Capacity of Asset to its Total Weight
	Assess Usage of Individual Assets	Load Factor	_	Utilized Capacity to Available Capacity Improvement Using Load planning
		Route Planning	_	Total Distance Traveled Total Lost Time in Traffic
	Assess Usage of Collective Assets	Avoiding Congestion, Providing More Direct Connections	_	Average Waiting/Idle Time of Vehicles/Vessels for Unloading/ Loading
		Expanding, Upgrading Infrastructure	_	Using Larger Vessels Avg. Turn-around Times in Ports/ Terminals
		Smart Traffic Management	_	Idle Time

Simulation is still the most common approach for energy saving calculations and lack of decision support systems especially for the supply chain with synchromodal capability. In the review no user-friendly tool was found that focuses on fuel consumption for multi-modal/synchromodal transportation considering all four modes of transportation – road, rail, air, and water (inland and sea). All of the reviewed tools focused on specific routes and didn't consider the whole supply chain. These tools were not comparison based and focused on calculating CO_2 emissions. Changing environmental impact to monetary value was also not found in the tools.

In the next section, all of the research gaps found in the literature are presented, together with how this study addresses them.

2.3. Literature summary and associated research gaps

In recent decades, although reduction of carbon emissions is recommended and sometimes enforced by law, cost reduction still is the primary motivation for business and supply chain owners. If being greener could be tied to spending less, then supply chains will be motivated to shift to better options. Meanwhile, suggesting drastic changes and configurations aimed at developing a greener supply chain does not help supply chain owners take advantage of small changes which add profit and sustainability.

Since the configuration of a supply chain is a strategic decision, having a rough estimation that gives the opportunity for instant comparison to the decision maker with an alternate method is a more useful approach than giving a final CO_2 emission total, which is what the reviewed software and associated algorithms provide. Unfortunately, because of the high complexity and many variables in energy calculation and fuel consumption within a supply chain, efforts to help supply chain owners see the monetary outcome of being greener fall short. Few tools can determine carbon footprints based on the configuration of a supply chain (Cholette and Venkat, 2009).

If accounting and logistics professionals would like to guide supply chains to address environmental issues, then they need more user-friendly tools. In this competitive economic world, not all decision makers feel secure using online tools to make strategic decisions and would like to be able to have a tool customizable to their needs, without sacrificing awareness of a tools assumptions. Customization by the decision maker gives them the opportunity to manipulate the use of different options instantly and observe the effects. The lack of such decision support systems was observed in the above literature review.

This paper presents and offers a tool for analysis of fuel consumption factors within a supply chain. The following is a summary of literature gaps found which SySCEA addresses:

- No tool was found that covers all four modes of transportation in a decision support system and could apply the synchromodality concept for estimation of energy cost and environmental effects in supply chains.
- The quantitative effect of customer orders and demands on supply chain energy cost has not been discussed. Our comparison-base tool considers customers' order-and-demand effect on total fuel consumption and shows how by changing the order quantity and yearly demand, the total fuel consumption and eventually CO_2 emissions will change.
- SySCEA focuses on calculation of fuel consumption in transportation rather than considering overall energy use. This helps supply chain staff responsible for ordering to make more precise managerial decisions by observing in detail the cost of transportation.
- Modeling within the reviewed literature was scenario-based. SySCEA can be used for any general supply chain under certain conditions.
- Partial loading is rarely considered in the literature, but SySCEA does incorporate partial loading which can make a difference in transportation costs, specifically for long-haul freight transportation.
- Simulation is the main method for estimating energy costs and traditionally it has been expensive and time-consuming. SySCEA uses mathematical modeling for calculation and is implemented in Excel as a user-friendly and low-price tool which is widely used by business.
- Parameters need to be flexible and easy to change by the user. Transportation assets, fuel types, and variations in packaging types are volatile, thus, within SySCEA the parameters can be customized by the user to easily fit the supply chain.

3. Research methodology

SySCEA is designed to explore the relationship between energy cost and consumption and the supply chain design and operation. The focus of the model is on energy cost and consumption because as energy consumption is reduced so are the greenhouse gas emissions, which results a smaller carbon footprint. In SySCEA the configuration, supply chain structure and energy efficient inventory policies are viewed through the lens of total fuel consumption. The new energy efficiency tool also considers partial loading for multimodal transportation and handles different parts with different packaging throughout the supply chain. Moreover, SySCEA gives the option of mode change to the user which is an inseparable part of future synchromodal networks.

The following are some assumptions that are considered throughout SySCEA development:

1. The investment that the supply chain makes to increase energy efficiency is not considered.
2. 2. The imperial measurement system is used.
3. Final results are reported in BTU, CO_2 emissions and dollars representing energy spent and saved.
4. Every transportation asset, route, and network is unique. For calculations, averaged numbers are used and the most general or standard situations are considered, while exceptions are excluded.
5. SySCEA is more effective for multi-modal container-unit cargo, but can be used for other types of freight as well.
6. Each transportation asset is restricted to carrying freight and sharing freight. Passengers are not considered.
7. Annual demand can change, but per period order quantity is fixed throughout the year.

Table 3
Top factors for supply chain energy savings.

#	Decision Factor	Impact Area
1	Number of distributions centers/local warehouses	Distance
2	Free space/gap in box/pallet/container	Load Weight and Volume
3	Lead times	Order Quantity
4	Total capacity of each asset/containers	Asset's Energy Efficiency
5	Stackability of packages	Asset's Energy Efficiency
6	Average distance between factories and distribution centers/distribution centers and customers	Distance
7	Efficiency of energy usage	Asset's Energy Efficiency
8	Improvement using load planning	Asset's Energy Efficiency

8. SySCEA gives an estimation of transit time in each route and the total for the whole supply chain. This estimation is roughly based on historical data averages and can be adjusted based on the user's preferences.

To utilize SySCEA, shipment data is entered by the user, together with air, water, road, and rail transportation costs, to estimate the annual cost of products shipped from suppliers to the customer.

In the very first step of modeling, the first five main categories of energy saving areas within a supply chain are introduced (Fig. 4). The categories are:

1. Increased value density
2. Reduced average transportation distance
3. Energy efficient inventory policies
4. Energy efficient asset technologies
5. Energy efficient usage of individual/collective assets

These categories are then broken into smaller quantifiable variables and a conceptual model is developed. Due to the high complexity of transportation systems in supply chains, a system dynamics approach is used to recognize the behavior of the system. Fig. 4 presents a cause and effect diagram of energy savings in the supply chain. The positive and negative signs on the arrows show the impact of variables on energy savings. This diagram is used to conceptually model the problem.

4. Mathematical modeling

The mathematical model was developed based on a set of conceptual model variables. These variables are categorized and described in Table 2.

4.1. The most significant variables

According to the Pareto principle, not all 28 factors have the same impact on supply chain efficiency. To find the most significant variables in supply chain energy saving, a group of experts in CELDi[2] was asked to fill out a survey and pick the top factors based on their experience in their field. The top factors chosen are provided in Table 3.

In the next step, we analyzed selected factors to formulate the problem. The key to linking these variables to calculate the total annual fuel consumption of orders in a supply chain is defining a suitable measurement unit. The result is very sensitive to this selection. To quantify the selected factors and calculate the energy consumption within the supply chain three main entities of a

[2] The CELDI industrial advisory board is comprised of supply chain logistic managers from a wide range of different industries.

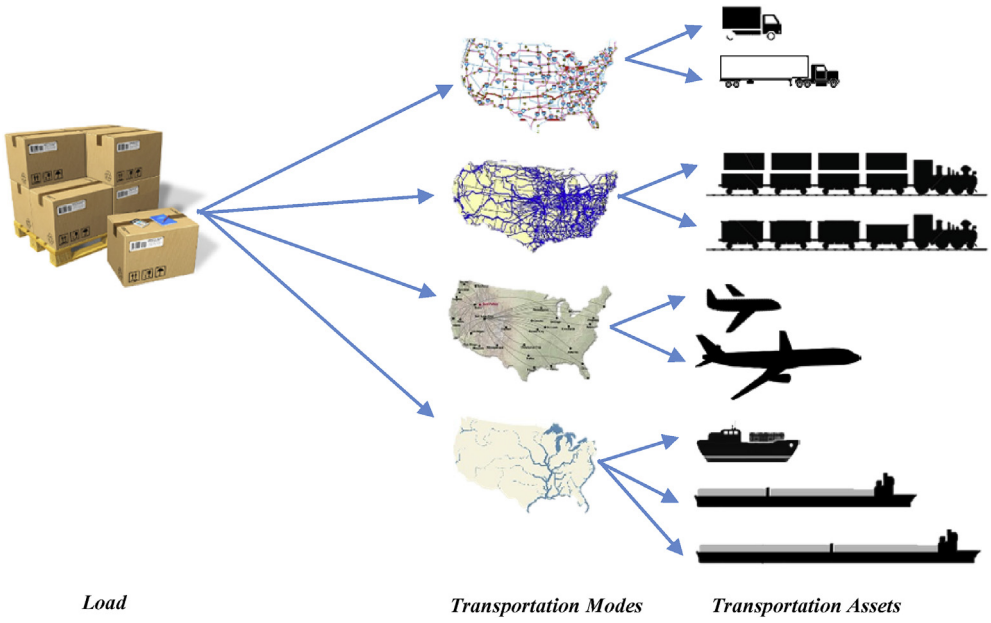


Fig. 5. A freight transportation schema in a supply chain.

Table 4
Interrelation of energy consumer entities and energy saving factors in a supply chain.

Factors and Variables	Load	Transportation Mode	Transportation Assets
Energy consumer areas	**Weight** of load	**Distance** traveled by loaded asset	**Fuel usage** during travel
Energy saving areas	More weight per volume	Less distance traveled	More energy efficient asset choice
Suggested measurement unit	Pound	Miles	Gallons/1000 ton. Miles

Based on these entities, the total energy used within a supply chain can be calculated.

supply chain that consume energy were found which can cover all other selected factors of energy saving. These entities are load, transportation modes, and transportation assets. (Fig. 5).

The relationship between these three entities (load, transportation modes, and assets), plus the selected factors with energy saving in the supply chain, can be seen in Table 4.

4.2. Formulation

In this section, the mathematical model and the calculations within SySCEA are discussed. The overall framework is shown in Fig. 6 and the numbers next to each object are used for referencing the various components throughout the following discussion. The overall objective is to calculate the total cost of energy in the transportation of a general supply chain and compare it with possible alternatives. To obtain the overall cost and conduct alternative comparison, the calculation of distances, load weight, energy efficiency of the assets, total supply chain energy measurements, alternative networks and associated parameters are needed as will be explained in detail in the following.

The following indices were used for formulation.

Indices :

i : node of origin
j : nodes of destination
p : type of products
m : mode of transportation
f : fuel type for mode of transportation
a : asset type for mode of transportation
t : transit time
S : speed
P : number of pallets
C : number of containers
A : number of assets
T : number of trips
V : volume
W : weight

Parameters :

Q_{ijpm} : order quantity of *product p* transported from *i* to *j* using *mode m*
D_{ijpm} : distance between *i* and *j* in the network traveled by *product p* in *mode m*
Dt_{ijmp} : dwell time between *i* and *j* in the network by *mode m* for *product p*
S_m : Average speed of traveling in *mode m*
K_{ijpm} : annual demand of the *product p* transported from *i* and *j* in the network by *mode m*
W_e : empty weight
W_{Max} : max weight
F_{ijmpf} : fuel efficiency of *asset a* in *mode m* by *fuel f*
V_{ijmp} : volume of order carried in between *i* and *j* in the network by *mode m* for *product p*
C_{ijf} : carbon emission index for *arc ij* of *fuel f*
E_{ijf} : energy consumption index for *arc ij* of *fuel f*
$F_{\$}$: fuel price of the selected asset of *arc ij*

Varibles :

W_{ijmp} : weight of order carried in between *i* and *j* in the network by *mode m* for *product p*

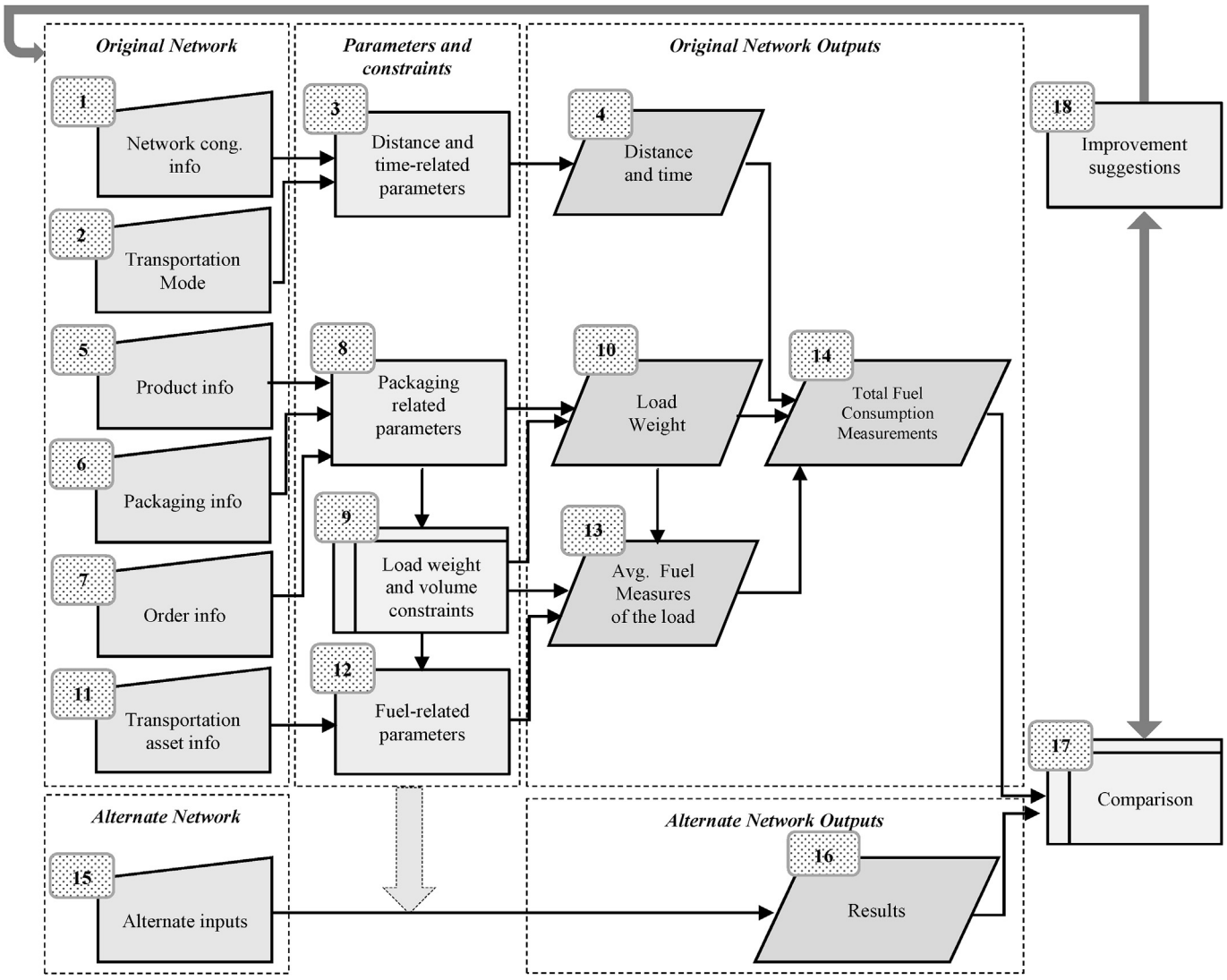


Fig. 6. SySCEA framework.

t_{ijmp} : Transit time between *i* and *j* in the network by *mode m* for *product p*
W_a : asset weight (including the load weight)
N_p : orders per year for *product p*
A_{ijmp} : number of asset used to carry load from *i* to *j* in the network by *mode m* for *product p*
FC_{ijmpf} : fuel consumption of *arc ij* in *mode m* to move *product p* using *fuel f*
TFC : total fuel consumption of the supply chain
CE_{ijmpf} : carbon emission of *arc ij* in *mode m* to move *product p* using *fuel f*
EC_{ijmpf} : energy consumption of *arc ij* in *mode m* to move *product p* using *fuel f*

4.2.1. Specifying parameters

To gather data for SySCEA development and future parameter settings, a literature review was conducted. In the case studies found at most three modes were considered simultaneously. Table 5 provides an overview of the modes discussed in the literature reviewed.

There are groups of parameters used in SySCEA which can be categorized into different subsections: distance and time-related, packaging-related and fuel-related parameters.

4.2.1.1. Distance and time-related parameters. Fig. 6, Item 3, contains the distance and time-related parameters required to determine distances between different origins and destinations. Four modes of transportation, road, rail, water, and air were considered for freight movement. For each mode of transportation, GIS maps were used to extract the shortest path between the nodes using the network analysis option of Esri's ArcGIS map 10.2. Maps sources were from Esri and the U.S. Department of Transportation.

A distance table was created using 114 cities from the lower 48 states. These cities were selected based on the most common freight terminals within the United States. Using the from-to combinations of the 114 cities a table was created that has 12,329 records.

From an accessibility point of view, not all 114 cities are covered by all four modes of transportation. Roads are the most common routes for freight movement and has the most coverage. All 114 cities have road access to each other. However, all paths and arcs are also covered by air transport. Freight rail almost fully covers the U.S. except for a few cities that are more than 25 miles away from any railroad. The user will see "N/A" in such cases when considering these paths for rail freight movement. For waterways, both inland waterways and intercostal waterways are considered. Again, not all 114 cities have access to waterways. The first/last 50 miles of

Table 5
Transportation modes considered in the literature.

References	Transportation Mode				Objective	Description	Notes
	Road	Rail	Air	Water			
Mohammadi et al. (2017)	✓	✓			minimizes total risk of shipments	Modeling a multi-modal hazardous materials transportation network under uncertainty	Defines the parameters needed for multimodal network model
Kapetanis et al. (2016)	✓	✓		✓	Calculates emissions, cost and time	An analytical fully parameterized model for one case study of Shanghai–Piraeus–Prague	It focuses on full load 20′ container for a certain route and is used for validation of the results
Li et al. (2017)	✓	✓		✓	Minimizing total delivery cost	Cooperative synchromodal freight transport planning among multiple transport operators	Simulation based and it was used for conceptual modeling of the problem
O'Kelly (2014)			✓		Total fuel used over the supply chain	A data based analysis of fuel usage in air freight	This data used for airfreight fuel cost estimation
Tolliver et al. (2013b)	✓	✓		✓	Rail fuel efficiency rates calculation	Models railroad energy efficiency models for different classes of trains and commodities	This study used for the US train fuel efficiency calculation also it compares rail to road and inland waterway
Davis et al. (2013)	✓				Fuel efficiency measures calculation	U.S. highway transportation energy efficiency of different vehicles	This study used for fuel efficiency estimation of road vehicles
Kruse et al. (2012)	✓	✓		✓	Emissions and fuel consumption measurement	Multimodal comparison study of freight fuel efficiency of assets in the US	This study used for comparison of the result and generating in land waterway measures
Institute for Water Resources (2012)	✓	✓		✓	Measurement of economic and environmental impacts of a chosen vessel	Post-Panamax vessel study	This study used for waterway measurement reference
Wigfross (2012)				✓	Total fuel consumption of different ships	Comparison of ships fuel efficiency	This study used for ships fuel efficiency estimation
Franzese and Davidson (2011)	✓				Total gallons of fuel and CO2 emission	Modeling the fuel consumption of tractor trailer class 8 under different condition of speed and grade	This paper was used as reference formula for tractor trailer class 8
American Bureau of Shipping (2011)				✓	Optimizing fuel consumption in ships	Fuel saving estimates for ships	This study used as a reference for fuel consumption of containerships
Becker (2011)				✓	Measuring efficiency of barges over Mississippi	Feasibility study of container transport over Mississippi river ending at port New Orleans	This study used as a reference for fuel consumption of barges
National Research Council (2010)	✓				Fuel consumption reduction calculation of trucks	Review of technologies and measurement of costs of fuel of trucks under different condition	This study was used for verification of measurement used for trucks
ICF (2009)	✓	✓			Calculation of fuel efficiency of truck and rail	Comparison of rail and truck fuel efficiency and consumption	This study used for comparison base of the result for rail and road
Notteboom and Cariou (2009)				✓	calculating the bunker cost for a specific service	Bunker costs and comparison of estimates on fuel costs	This study used for understanding the limitations of waterways transportation
Parajuli (2005)	✓	✓			Rail and road fuel consumption	Comparison of rail and road fuel efficiency	This study was used for verification of the estimates for rail
Office of Transportation and Air Quality (2000)				✓	Diesel-powered marine vessels emission measurement	an analysis of vessel classifications and horsepower to vessel weight relationships	This study was used for defining measures for vessel fuel efficiency

transportation from the origin/departure nodes will be marked as inland or coastal ports when available and the distance calculated. N/A will designate those nodes that are not available. The first and last mile transportation will be considered in the model later.

Finally, in order to estimate transportation time, the average speed of each mode and the expected dwell times are obtained from the user or default values from historical data. The speed and dwell time is used for the transit time calculation.

4.2.1.2. Packaging-related parameters. UPS and FedEx freight catalogs where chosen as the base to determine packing parameters. The most frequently used types of pallets and containers for product shipments are chosen for the packaging options at Item 8. These pallets and containers were also chosen for multimodal transportation. Interior and exterior dimensions, empty and max weight of pallets and containers are used for calculation of the load weight (see the Appendix).

4.2.1.3. Fuel-related parameters. Fuel-related parameters are divided into two categories, economic and environmental. For economic parameters, the fuel prices for different types are obtained from the user. For environmental parameters, CO_2 and BTU generation of each fuel type and the energy efficiency of each asset group are used. These parameters are in Items 12 of the framework and are used for calculating the average fuel of the load and the total fuel consumption for each route and the overall supply chain, respectively, in Items 13 and 14.

Fuel prices vary by mode type and their CO_2 emission and BTU generation rates also vary. Fuel prices can be updated by the user as an input parameter to determine the final cost of the fuel consumed in the supply chain. The user has the flexibility of adding a user defined fuel type. For BTU and CO_2 emissions, default values for the parameters are available for million BTU per gallon and pounds of CO_2 emissions per gallon. The user can overwrite/update the default values. Using these parameters, the total CO_2 emission and BTU generation for each arc is calculated and reported to the decision maker as shown in Item 13.

The energy efficiency of each asset was more challenging to determine. As was mentioned in the literature review section, no paper was found to tackle all modes of transportation simultaneously. The parameter setting of the tools used in previous research were inconsistent since every paper has its own assumptions. Every mode of transportation used different measures to determine its energy efficiency. In order to deal with mode changes this tool required a unique measure approach.

The steps below were followed to find the required parameters.

1 Study different assets of transportation modes and categorize them
2 Find the frequency of use for the assets and choose the most common ones in each category
3 Add fuel type, max payload, and volume capacity for the chosen assets

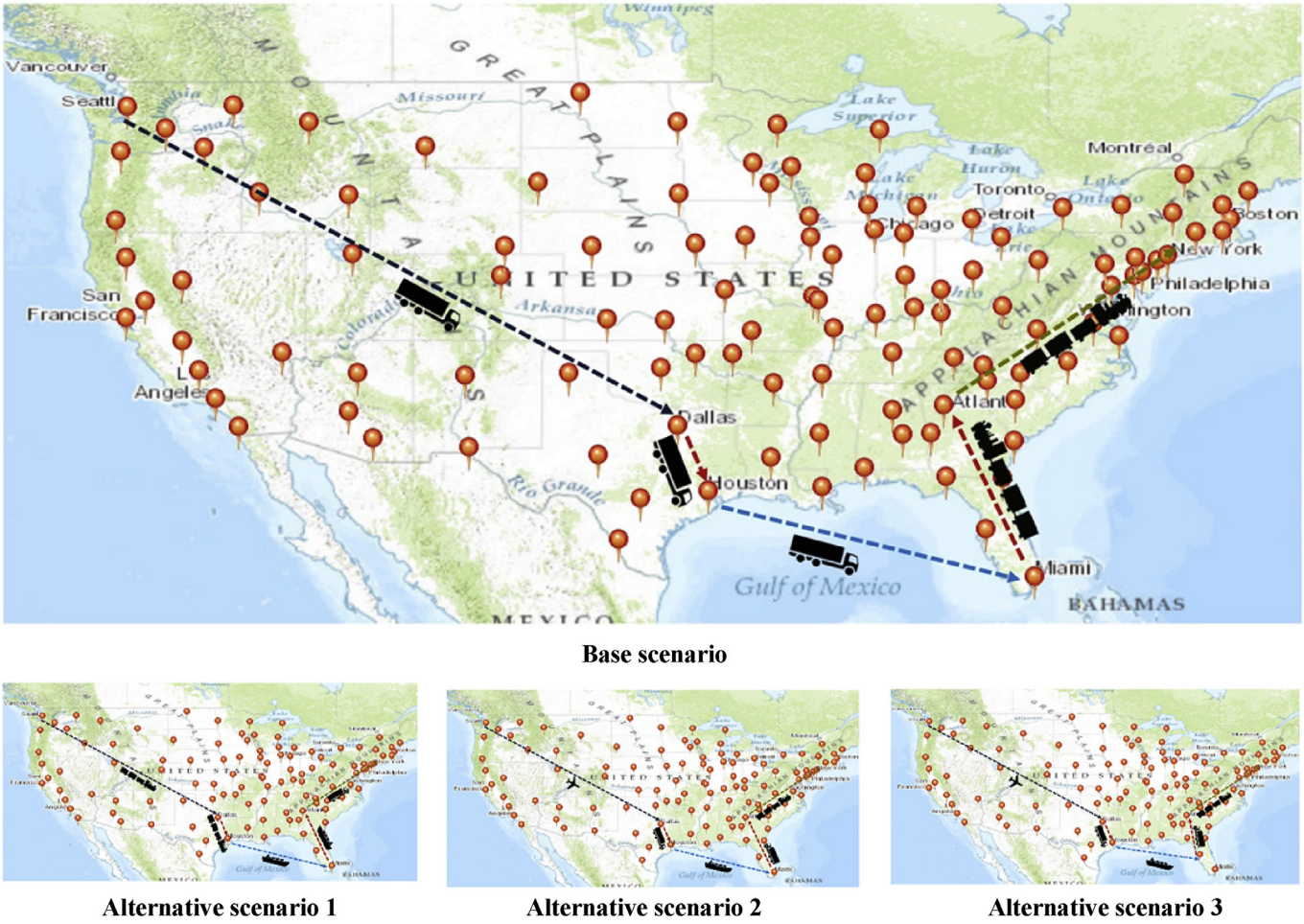


Fig. 7. Network configuration of the base and alternative scenarios.

4 Calculate the fuel efficiency index of the selected assets (gallons per thousand ton-mile)

To determine the asset list for each mode of transportation a study was conducted to find the most common transportation assets used in the U.S. for carrying freight. Using company catalogs and the annual reports of leading companies' production and sales, the largest category was chosen and the most common assets were selected. The National Academy of Sciences states that the fuel consumption of diesel and gasoline engines have the potential to be improved by up to 24% by changing their related variables (NRC, 2010; Vehicles, 2010). Therefore in addition to the most commonly used assets, the user has the option to choose among assets with either +10% and +20% fuel efficiency improvement to allow the decision maker the opportunity to upgrade transportation assets.

Calculation of energy consumption in a supply chain itself is complicated. Because so many factors affect fuel consumption and the estimation of CO_2 emission for transportation the problem needs to be simplified (Harris et al., 2011). In order to calculate fuel consumption, choosing the right measurement unit and trustworthy numbers are important. Different modes of transportation have different characteristics and their sensitivity to the load weight depends on the mode. Unfortunately, in the literature different numbers and measures are used which sometimes contradict. This resulted in the need to define a base for parameter recalculation.

To find the fuel efficiency, road transportation was examined since it tends to be one of the most sensitive modes of transportation (NRC, 2010; Vehicles, 2010). Miles per gallon (mpg) is the common unit for light-duty vehicles and has been used for fuel consumption in the literature (e.g. Harris et al.). However, load weight significantly impacts fuel consumption in road transportation. For example, for cars and trucks their range can be from 6.5 gallons per thousand ton-miles to 69 gal/1000 ton-miles due to changing the weight and size of the truck (Davis et al., 2013). Note that a fully loaded truck would have a much lower load-specific fuel consumption ratio than a partially loaded truck (NRC, 2010).

Another issue is the definition and calculation of the gallons per ton-miles index. The gallons per ton-mile index is calculated for a full load and a full fuel tank that travels the maximum possible distance. The fuel efficiency (gal/ton-mile) for moving the freight from node i to j can be calculated as

$$F_{ijmpf} = \frac{V_{\max A_{mijpf}}}{W_{\max A_{mijpf}} \cdot D_{\max A_{mijpf}}} \qquad \forall i,j \quad i \neq j \tag{1}$$

Although this measure has been used in literature, it is only applicable when a long-haul truck is fully loaded. So, for partially loaded trucks a more precise calculation is needed. To determine partial load fuel usage Franzese and Davidson (2011) built a model for average fuel efficiency (mpg) by using data for a vehicle weight between 20,000 and 80,000 lbs. by 10,000 lb. increments and 65 mph speed (the most common speed on freeways). Their data showed higher speeds decrease fuel efficiency and the vehicle weight relationship is nonlinear for vehicles above 65,000 lbs. Note that the asset weight consists of load weight and the empty asset weight. Equation (2) is interpolated based on actual data for fuel consumption.

$$F(W_a) = -5E-10W_a^2 \pm 6W_a + 9.6687 \tag{2}$$

In our tool, the formula of Franzese and Davidson (2011) and gal/ton-mile index are used by dividing Eq. (2) by load weight.

Eq. (2) is based on vehicle weight of 10,000 lbs. However, the majority of delivery trucks weigh more than 12,000 lbs. (Davis et al., 2013), therefore, Eq. (2) can be used for smaller size trucks to calculate a lower limit for fuel consumption for road transportation considering their different empty vehicle weights. The maximum load weight of the freight vehicle is used as a parameter to add the number of long-haul trucks required for freight transportation.

Apart from asset fuel efficiency, another issue to deal with is different fuel consumption for trucks with different loadings over a single route. For one order any number of fully loaded trucks and partially loaded trucks may be needed. The ability to calculate the average fuel efficiency for one route considering partially loaded trucks was needed. To deal with this problem, average weighted fuel consumption was defined (i.e., the average fuel efficiency of a truck is the weighted average of fuel efficiencies for fully loaded trucks and partially loaded trucks).

Trains and ships are not very sensitive to the load weight because their empty weight is quite heavy. Estimates from the literature and official reports were used as the basis for their fuel efficiency. For air transportation, the calculation was based on catalogs of airplane manufacturers. (See the Appendix for more information).

4.2.2. Network configuration, distances and transit time

As noted earlier, transportation modes and distances have a direct relationship with the fuel consumption of a supply chain. SySCEA has the flexibility of choosing the origin and destination nodes and allowing the mode of transport to show the actual distance used for calculation. To find the actual distances for each mode, an ArcGIS map for each mode is used, but the user can change the distance table and customize it for the supply chain network.

In Item 1 (Fig. 6), the user is first asked to enter the network configuration information. This is accomplished by breaking down the original network into arcs and entering the specifications of each route separately. For each arc, the transportation mode needs to be selected (Item 2). Items 1 and 2 are used for the calculation of arc distances in Item 3. (Refer to parameters section for further details of distance and time-related parameters).

The user can choose the nearest city to the desired O/D and the mode of transport which is used to calculate the actual transportation distance.

SySCEA also calculates the transit time for each chosen O/D pair using the formula below:

$$t_{ijmp} = \frac{D_{ijmp}}{S_m} + Dt_{ijmp} \tag{3}$$

4.2.3. Load weight

As mentioned in Table 2, one way to save energy is by reducing free gaps in packaging layers. This gap reduction affects both load weight and volume. Load weight is a primary variables used in the calculation of fuel consumption in a supply chain. Load weight (Item 10) is calculated using product, packaging, and order information considering load and weight parameters and constraints (Items 5, 6, 7, 8 and 9).

In the product information data entry, the user is asked to enter the weight and dimensions of the original box (package), order quantity and annual demand for the product. In the packaging information data entry, up to three layers of packaging are considered: the original box, pallets, and containers. For pallet and container types, a data sheet for the most common packaging specification in the U.S. are provided,i.e., exterior length, width and height, inside length, width and height, as well as empty and max weight of the layer in packaging related parameters at Item 8 (see the Appendix for more detail). In the input data sheet, the user enters the weight and size of the first box and chooses the second

Table 6
Definitions of four scenarios.

Inputs	Arc #	Base Scenario		Alternative Scenario 1	
Distance Data	1'	First mile (Origin→ Seattle)	Road	First mile (Origin→ Seattle)	Road
	1	Seattle → Dallas	Road	Seattle → Dallas	**Train**
	2	Dallas → Houston	Road	Dallas → Houston	**Train**
	3	Houston → Miami	Road	Houston → Miami	**Water**
	4	Miami → Atlanta	Train	Miami→ Atlanta	**Water**
	5	Atlanta → New York	Train	Atlanta → New York	**Road**
	5'	Last mile (NY → Destination)	Road	Last mile (NY → Destination)	Road
Order Data	1-5	Order quantity: 1000 units		Order quantity: 1000 units	
	1-5	Parts annual demand: 5000 units		Parts annual demand: 5000 units	
Packaging Data	1-5	Standard Pallet	40' container	Standard Pallet	**20' container**
Asset Data	1	Tractor Trailer	Diesel	**Double Stack car Train**	Diesel
	2	Tractor Trailer	Diesel	**Double Stack car Train**	Diesel
	3	Tractor Trailer	Diesel	**Barge carrying with max load capacity 2,000 TEU**	Diesel
	4	Flat Car Train	Diesel	**Barge carrying with max load capacity 2,000 TEU**	Diesel
	5	Flat Car Train	Diesel	**Tractor Trailer**	Diesel
Inputs	**Arc #**	Alternative Scenario 2		Alternative Scenario 3	
Distance Data	1'	First mile (Origin→ Seattle)	Road	First mile (Origin→ Seattle)	Road
	1	Seattle → Dallas	**Air**	Seattle → Dallas	**Train**
	2	Dallas → Houston	Road	Dallas → Houston	Road
	3	Houston → Miami	**Water**	Houston → **Tampa**	Road
	4	Miami → Atlanta	**Road**	**Tampa** → Atlanta	**Road**
	5	Atlanta → New York	Train	Atlanta → New York	**Air**
	5'	Last mile (NY → Destination)	Road	Last mile (NY → Destination)	Road
Order Data	1-5	Order quantity: **4500 units**		Order quantity: 1000 units	
	1-5	Parts annual demand: 5000 units		Parts annual demand: 5000 units	
Packaging Data	1-5	Standard Pallet	**20' container**	**Paper-Base Pallet**	**20' container**
Asset Data	1	**Airbus A300**	**Jet Fuel**	**Flat Car Train**	Diesel
	2	Tractor Trailer	Diesel	**Tractor Trailer with 20% better fuel efficiency**	Diesel
	3	**Barge with 2000 TEU**	Diesel	**Tractor Trailer with 20% better fuel efficiency**	Diesel
	4	**Tractor Trailer**	Diesel	**Tractor Trailer with 20% better fuel efficiency**	Diesel
	5	Flat Car Train	Diesel	**Boeing 757**	**Jet fuel**

and third layer, if any, from the drop-down list of the most common pallets and containers. If the pallet or container is customized and is not on the list, there is an option for "user-defined" pallets or containers in the data sheet. Once the data is entered, the maximum possible number of pallets and containers is calculated considering the weight and volume constraints (Item 9) using Eqs. (4) and (5). The user has the flexibility to overwrite the proposed number of pallets and containers.

The load weight and volume constraints (Item 9) are conceptually given below.

Constraint 1 : *total weight of load per pallet, container and asset* $\leq$ *maximum load of pallet, container and asset*

Constraint 2 : *total volume/area of load per pallet, container and asset* $\leq$ *maximum volume/area of pallet, container and asset*

Equations (4) and (7) calculate the suggested number of pallets and containers for fully utilized pallets and containers and equations (5) and (8) calculate the suggested number of pallets and containers for partially utilized pallets and containers. Equations (6) and (9) provide the options that are given to the decision maker for quantities of pallets and containers. The default option is the minimum number of pallets and containers.

Finally, the load weight is determined using Eq. (10) based on the weight of an empty pallet, empty container, and product given their quantities per load.

$$P_1 = \frac{Q_{ijpm}}{MIN\left(\frac{W_{Max\ p} - W_{e\ p}}{W_p}, \frac{V_p}{V_p}\right)} \quad \forall i,j\ i \neq j \tag{4}$$

$$P_2 = \frac{Q_{ijpm}}{MIN\left(\frac{W_{Max\ p} - W_{e\ p}}{W_p}, \frac{V_p}{V_p}, \frac{A_p}{A_p}\right)} \quad \forall i,j\ i \neq j \tag{5}$$

Table 7
Results of four scenarios and their comparison.

Status		Arcs							
		First mile	Arc one	Arc two	Arc three	Arc four	Arc five	Last mile	Total supply chain
Base Scenario	Fuel cost ($)	56	3826	415	2063	312	396	326	7394
	Transit time (h)	1.2	39	5	23	54	62	2	187
Alt. Scenario 1	Fuel cost ($)	282	641	72	216	145	1532	65	2954
	Transit time (h)	1	118	35	97	73	17	0.2	342
Alt. Scenario 2	Fuel cost ($)	421	6791	429	169	1211	256	976	421
	Transit time (h)	0.5	7	5	97	13	62	0.9	186
Alt. Scenario 3	Fuel cost ($)	282	819	1250	5142	2474	4624	160	14,750
	Transit time (h)	1	118	5	19	10	5	0.5	159
$ saving by Alt 1 (%)		−80.0%	83.2%	82.7%	89.5%	53.4%	−287.0%	400.0%	60.1%
Time saving by Alt 1 (%)		23.1%	−200.5%	−562.8%	−326.7%	−35.6%	72.1%	90.5%	−83.5%
Changes in Alt 1		Different origins	Shifting to train	Shifting to train	Shifting to barge	Shifting to barge	Shifting to truck	Different end points	
$ saving by Alt 2 (%)		−647.6%	−77.5%	−3.4%	91.8%	−287.7%	35.4%	−199.1%	−38.6%
Time saving by Alt 2 (%)		61.5%	81.2%	0.0%	−326.7%	75.5%	0.0%	110.0%	0.4%
Changes in Alt 2		Different origins	Shifting to air	Change of order plan and container	Shifting to containership	Shifting from train to road	Change of order plan and container	Different end points	
$ saving by Alt 3 (%)		−400.0%	78.6%	−201.5%	−149.3%	−525.1%	−1317.4%	104.3%	−99.5%
Time saving by Alt 3 (%)		23.1%	−200.5%	0.0%	16.6%	82.3%	91.2%	76.2%	14.7%
Changes in Alt 3		Different origins	Shifting to train and paper pallets	Shifting to energy efficient vehicle and paper pallet	Rerouting, energy efficient vehicle and paper pallets	Rerouting, energy efficient vehicle and paper pallets	Shifting to air and larger plane	Different end points	

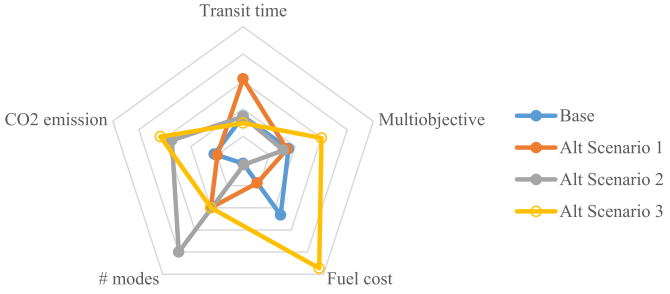


Fig. 8. Comparison of scenarios based on different criteria using spider plot.

$$P_2 = \frac{Q_{ijpm}}{MIN\left(\frac{W_{Max\ P}-W_{e\ P}}{W_p}, \frac{V_P}{V_p}, \frac{A_P}{A_p}\right)} \quad \forall i,j\ \ i \neq j \tag{6}$$

$$C_1 = \frac{P^*}{MIN\left(\frac{W_{Max\ C}-W_{empty\ C}}{\left(MIN\left(\frac{W_{Max\ P}-W_{empty\ P}}{W_p}, \frac{V_P}{V_p}\right)\cdot W_p + W_{empty\ P}\right)}, \frac{V_C}{V_P}\right)} \quad \forall i,j\ \ i \neq j \tag{7}$$

$$C_2 = \frac{P^*}{MIN\left(\frac{W_{Max\ C}-W_{empty\ C}}{\left(MIN\left(\frac{W_{Max\ P}-W_{empty\ P}}{W_p}, \frac{V_P}{V_p}\right)\cdot W_p + W_{empty\ P}\right)}, \frac{V_C}{V_P}, \frac{A_C}{A_P}\right)} \quad \forall i,j\ \ i \neq j \tag{8}$$

$$C^* = \{C_1,\ C_2,\ MIN(C_1, C_2),\ user\ preferece\} \tag{9}$$

$$W_{ijmp} = (C^*.W_{empty\ C}) + (P^* \cdot W_{empty\ P}) + Q_{ijpm} \cdot W_p \quad \forall i,j\ \ i \neq j \tag{10}$$

Note that the interior and exterior volume/area for pallets and containers are different. The default value is provided, but the user has the option of overwriting with different values. In such cases, the calculation is done with user inputs. A single product type per load is the base assumption for the calculation of the total net weight of the freight. If the freight consists of more than one product and the total weight is known, simply substitute this weight for the total weight of the products and the rest of calculation will not change. Also, if the products are already palletized or containerized, then the weight and number of pallets or containers can be added manually.

4.2.4. Fuel consumption measurements

After calculation of the distances and the load weight of Items 4 and 10, the energy efficiency of the selected asset is calculated in Item 13 of Fig. 6, using the transportation information from Item 11 and fuel-related parameters of Item 12.

In Item 11, the user is asked to choose the asset type from the drop-down box—see the Appendix for the options in each mode—and the fuel type. The energy efficiency of the selected asset is calculated using the load weight and asset parameters in the fuel-related parameters of Item 12. Then the average fuel consumption of each route is calculated as (Item 13):

$$FC_{ijmpf} = D_{ijmp} \cdot W_{ijmp} \cdot FE_{ijmpf} \cdot A_{ijmpf} \cdot \frac{K_{ijmp}}{Q_{ijmp}} \quad \forall i,j\ \ i \neq j \tag{11}$$

The average fuel consumption displayed in SySCEA is across all five arcs. However, that can easily be extended as the calculation of each arc is done separately and they add up to show the total fuel consumption for the defined supply chain. The user has the option of choosing different fuel types. There are several fuels types, but the main categories considered are gasoline, diesel and jet fuel. However, the user can update the numbers based on the supply chain and add more fuel types in the user-specified fuel type designation (Item 12). For each fuel type, the carbon emission and energy consumption parameters are also defined (see the Appendix). Using these parameters, the carbon emission and energy consumption for each route are calculated. Since decision makers often prefer comparison in monetary terms, it is possible to have the dollar value of fuel used in each arc and the sum of all arcs in the supply chain provided (Item 14, Fig. 6):

$$FC_{\$} = \sum_{i=1}^{\#Arcs} \sum_{j=1}^{\#Arcs} \sum_{m=1}^{\#\ modes} F\$_{ijmpf} \cdot D_{ijpm} \cdot W_{ijmp} \cdot F_{ijmpf} \cdot A_{ijmpf} \cdot \frac{K_{ijpm}}{Q_{ijpm}} \tag{12}$$

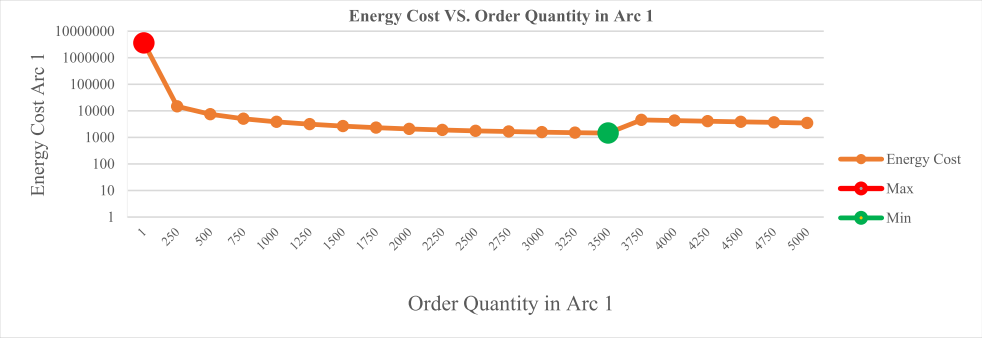


Fig. 9. Sensitivity graph of order quantity vs. energy cost of a supply chain.

4.2.5. Alternative network option

When entering the data for the current arc, it is possible to enter data for the alternate arc by using the same data in Item 15, with the results in Item 16. Then by changing alternate arc factors, the user can compare the results on the same page of the tool with the original network (Item 17). The alternative network analysis is the same as the original network, but gives the user the option of manipulating all inputs to see how the final results might change the percentage of dollars saved or lost.

4.2.6. Comparisons and improvement suggestions

As mentioned in the previous section, the user can see the final results of the original and alternative network together and compare them using Item 17. Multiple energy saving areas are considered including:

- Energy saving by configuration of the supply chain
- Energy saving by reduction of packaging weight and volume
- Energy saving by changing the order planning
- Energy saving by shifting in between the modes of transportation, assets and fuel types (if possible)

Referring to Items 14 and 16, the total fuel value consumption, CO_2 emission, BTU generation for either original or alternative arcs, and the comparison for savings/loss percentage gives the decision maker enough information to choose between saving areas and determine improvement suggestions in Item 18 for the overall supply chain. The decision maker has the flexibility to manipulate the different combination saving areas and see the influence on final measures to pick the most applicable ones for the supply chain. SySCEA also allows for graphical visualization of results. Product ordering can be a dynamic decision, therefore it is possible to view a sensitivity graph for the order quantity that shows the energy cost at different order sizes.

5. Tool application

Based on the cost and performance models developed, an Excel-based tool was created to find the relationship between energy cost/consumption and supply chain design/operation. SySCEA supports a supply chain with a maximum of five arcs (six cities/nodes with five routes between them) and can be used for any supply chain.

The user enters the requested data, input parameters for fuel types, speed, dwell time, cities and mode of transportation, ordering information, packaging information and asset information for each arc. Calculation for each arc is performed in separate Excel worksheets; however, the user can see the overall results simultaneously for verification. Additionally, the user has the option of changing input parameters to investigate the fuel consumption improvement potential for each arc. The cost of fuel is presented for each arc and for the total supply chain at the top of each arc input sheet. The total fuel consumption, the energy usage and the CO_2 emission for each arc are shown at the bottom of each arc input sheet.

The following will demonstrate how to use SySCEA. A base scenario and three alternatives were developed and their results are presented and discussed. Fig. 7 illustrates for the base scenario how the freight is passing through five arcs to get to the final destination and each arch uses a different mode of transportation.

Table 6 shows the inputs provided by the user. The base scenario and the alternatives are different regarding network configuration, mode of transportation, order quantity, pallet size, container size, asset and fuel type. The input data changed in each alternative scenario is shown in a bold font.

In Table 7, the results for all the scenarios and their comparison are given. For example, comparison of the base scenario with alternative 1 shows how shifting to more energy efficient decisions can save the supply chain approximately 60 percent of the original fuel cost. The savings come from changing to assets (carrier vehicles) and modes of transportation that are more energy efficient. The order size and container type effect depends on the carrier payload, but fully loaded long-haul trucks (weight-wise) are more energy efficient.

Fig. 8 shows how the different performance criteria for the four scenarios can be compared. Alternative 2 has the minimum fuel cost and transit time, but its CO_2 emission is as high as Alternative 3.

The Base scenario – rail and road - and Alternative 1 – rail, water, and road - are environmental friendly approaches. However, Alternative 2, which is a combination all modes, is the cost-efficient. Also by combining three objectives of transit time, fuel cost and CO_2 emission with equal weights it can be seen that Alternative 3, which is the combination air, rail and road modes, is not as good as others. So the decision maker can choose which scenario fits the supply chain performance requirements the best. For example, if time is a concern, SySCEA would not recommend Alternative 1. However, if it is desired to pick an environmental friendly option, then picking the Base scenario, which is a combination of road and rail, would be desireable.

Order quantity is the most challenging factor and it depends on multiple parameters of weight, volume, mode choice and asset type. For this factor SySCEA assist the decision making process by providing a sensitivity graph for each Arc. For example, for the Base scenario, Fig. 9 shows how Arc 1 total energy cost changes by changing the order quantity. The user can conclude from this graph that an order size of 3500 units is the most energy efficient size for Arc 1 with current freight size and other parameters.

There are other graphs in SySCEA to help the decision maker compare results. For each arc and its alternative scenarios, a chart is

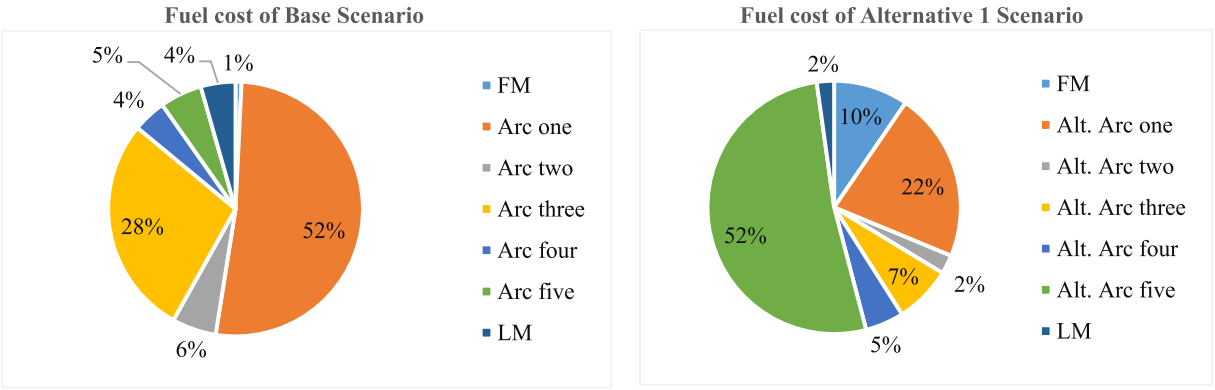


Fig. 10. % Share of arcs on Base scenario and Alternative 1 in overall fuel cost of the supply chain.

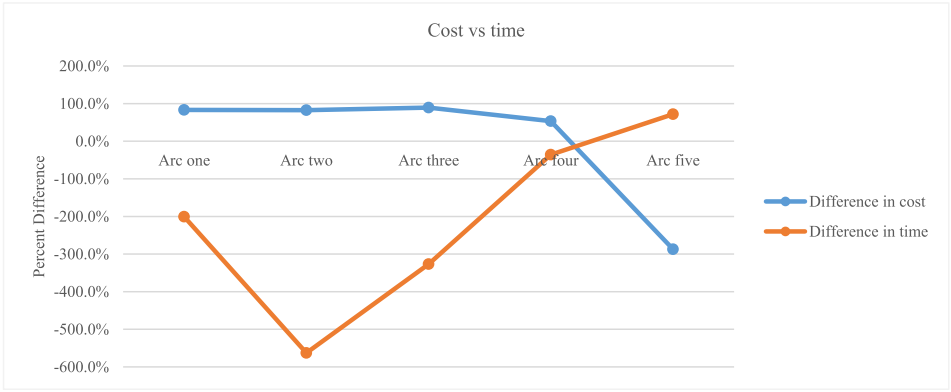


Fig. 11. % Difference (saving) for Alternative 1 in comparison with base scenario.

Table 8
Selected O-D pairs and comparison of the results to other tools.

O-D	Description							
	Source	Suggested modes	Distance (mile)	Time of selected tools (day)	Time for SySCEA (day)	CO2 emission of selected tools (lbs)	CO2 emission for SySCEA (lbs)	% Difference
Miami to Los angels	Alumur et al. (2012)	Truck-Air-Truck	2741	–	0.5	246,585	261,305	−6.0%
		Sea Freight	4951	–	12.5	7890	7562	4.2%
Atlanta to Houston		Air Freight	1430	–	0.25	71703	77947	−8.7%
Piraeus–Prague	Kapetanis et al. (2016)	Rail	954	2.29–5	2.7	64,815	62,014	4.3%
Shanghai–Piraeus		Sea Freight	9085	22.6	24	73,193	75,129	−2.6%
						Average of differences	**−1.8%**	

shown that displays the share of each arc in the total fuel cost of a supply chain. For example in Fig. 10, Arc 1 and Arc 5 consume the most fuel for the Base scenario and Alternative 1. Arc 1 contributes 52% of the supply chain's transportation cost, which uses road transportation, and Arc 5 in Alternative 1 is using 52% of the overall cost which also uses road transportation. Note that the Base scenario is using rail and road, but Alternative 1 uses rail, road, and water.

Finally, SySCEA provides a visualization of the changes in the fuel cost for each arc according to the scenario definition. For instance, Fig. 11 depicts that shifting from train to road can save 70% of time in Alternative 1 compared to the Base scenario, but the cost of fuel is almost tripled.

6. Comparison to existing tools

As discussed in section 2.2 of the literature review, we could not find any tool or paper that fully covers all dimensions that are included in this study. SySCEA is unique in its ability to support the different dynamics of a synchromodal supply chain. Synchromodal supply chains are lacking comprehensive decision-making tools to support their design. Therefore, to validate our results, we are limited to a partial comparison of the tools outcomes with existing tools or papers which are focused on single routes, rather than the whole supply chain, and with significant assumptions.

The online Emission Calculator tool (Multimodal Company, 2017) and the numeric results of a study by Kapetanis et al. (2016) are chosen for comparison. Both models have limited parameter settings, so SySCEA is adjusted based on the comparison model parameters to be able to obtain valid results. Both of these models cover a small set of selected O-D pairs and only provide the shortest path as the final route.

It can be seen in Table 8 that the average difference of the results for the chosen routes between SySCEA and selected models is less than 2%. It is valuable to note that CO_2 emissions are highly related to the type of fuel. These comparison studies do not address the

type of the fuel used, so the difference in the results might not be accurate due to different methods of calculation.

7. Conclusions and future work

Synchromodal supply chains focus on determining the best transport alternative in a multimodal network based on cost reduction, on-time delivery, and emission reduction, as well as improving the robustness and accessibility of a network. It is common for industry to be interested in knowing to what extent one supply chain decision is better than other alternative and comparison-based tools enable efficient analysis.

This paper explicitly investigated energy cost and consumption for synchromodal supply chain logistics by varying the most significant variables. The model was developed to handle both energy efficiency issues and supply chain design parameters to obtain new insights that support decision making under different scenarios.

SySCEA is an Excel-based model that serves as a tool to enable supply chain decision makers to perform the following tasks:

- Estimate the overall supply chain cost as related energy efficiency variables change.
- Evaluate the overall energy efficiency of a supply chain and determine how it might be improved.

SySCEA factors in energy costs and estimates fuel consumption and transit time. Decision variables include suppliers and vendors, warehouse locations, inventory policy (i.e., order quantity and annual demand), as well as mode choice and asset type across the supply chain. SySCEA provides results for the total value of fuel consumption in dollars, CO_2 emissions in pounds, transportation energy usages in BTU and transit time in hours. It enables comparative results and offers potential improvement strategies by considering other design/operating options such as reconfiguration of the network and changing the product size and volume, as well as the packaging characteristics and different transportation modes and assets. The tool is able to be customized by the user and can be used for general supply chains.

Future work will address the limitation of SySCEA's one-mode of transportation per arc or route assumption. Applying parallel multimodal transportation for a single route - modal share - is a planned extension of this model. This model does not investigate the trade-offs between the required investment and energy efficiency improvements. Adding this feature to SySCEA will help decision makers to plan more precisely for a greener supply chain from a monetary point of view.

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Appendix

1. Survey data

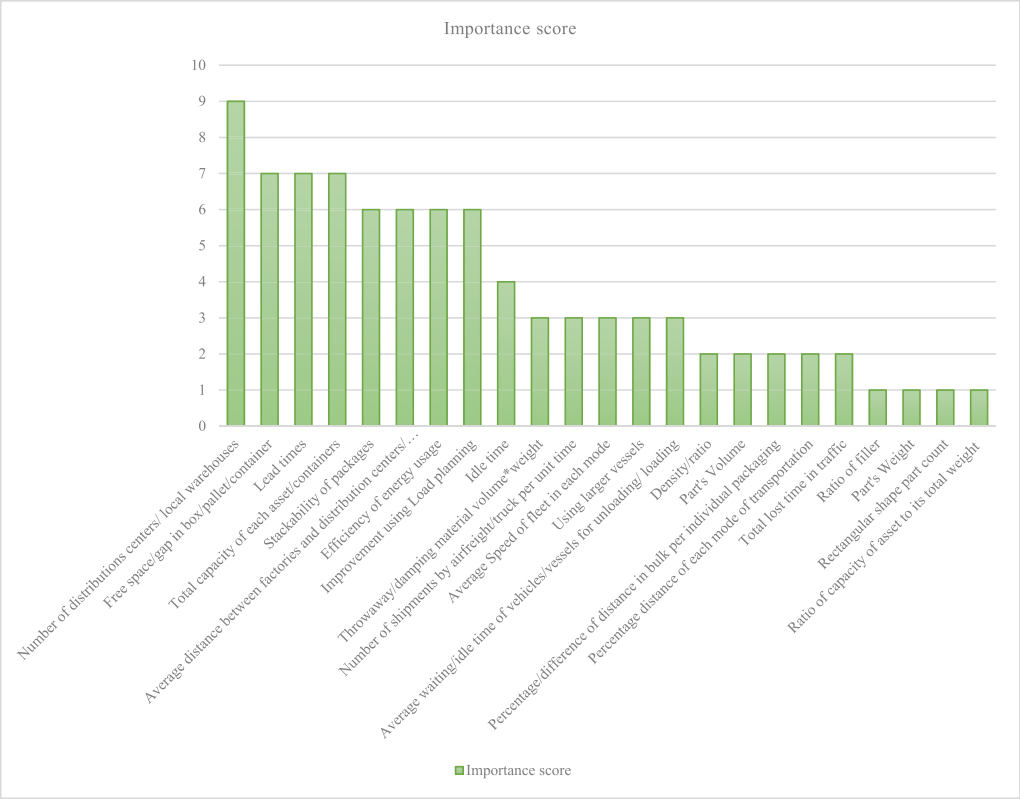


Fig. 1A. Importance score for the energy consumer factor.

2. Parameters

2.1. Packaging parameters

2.1.1. Pallets' sizes.

Table A 1
Pallet sizes

Pallet types	Exterior L	Exterior W	Exterior H	Empty weight	Max weight	Inside L	Inside W	Inside H
Standard Pallet (United Parcel Service of America Inc., 2016)	40	48	86	50	2000	39	47	80
Wood-Base Pallet (Pallet International Inc., 2011)	40	48	86	10	1500	39	47	80
EUR Pallet (Greenway Products & Services, 2016)	32	48	86	30	2000	31	47	80
Automotive Industry Pallet	45	48	86	70	2000	44	47	80
User Specified Pallet	96	61	64	183	2400	95	60	58
No Pallet	0	0	0	0	0	0	0	0

2.1.2. Container sizes.

Table A2
Container sizes (C.H. Robinson Worldwide, 2015)

Container types	Exterior L	Exterior W	Exterior H	Empty weight	Max weight	Inside L	Inside W	Inside H
20′	20	8	8.5	4850	66139	18.6	7.6	7.75
40′	40	8	8.5	8380	66139	39.5	7.6	7.75
40′ High	40	8	9.5	8598	58598	39.33	7.58	8.75
45′ High	45	8	9.5	10580	62970	44.33	7.6	8.75
48 High	48	8.5	9.5	10850	56350	47.5	8.5	8.83
53′ High	53	8.5	9.5	11100	56090	52.3	8.16	8.83
User Specified Container	40	8	8	8380	66139	39.5	7.6	7.6

2.2. Economic and environmental fuel parameters

2.2.1. Fuel Indexes.

Table A3
Fuel indexes (Energy Information Administration)

Fuel types	Price per gallon ($)	Million Btu per gallon	lb. of CO2 emission per gallon
Gasoline	2.22	0.1140	19.40
Diesel	2.57	0.1389	22.20
Jet fuel	1.86	0.1281	29.02
User specified	2.32	0.1500	24.84

2.2.2. Delivery or freight vehicle parameters.

Table A4
Freight vehicles (NRC, 2010; Vehicles, 2010)

Vehicle type	Max load	Min GVW	Max GVW
Delivery truck with max capacity of 11,000 lb.	11,000	12,000	23,000
Tractor-trailer (Class 8) with max capacity of 54,000 lb.	60,000	20,000	80,000

2.2.3. Rail car parameters.

Table A5
Boxcar sizes and their fuel consumption (Tolliver et al., 2013a, ICF, 2009, Sonesson and Berlin)

Category	Type	Fuel types	Actual weight (w/out freight) lbs.	Cargo weight/pay-load capacity max	Max Cargo volume (ft^3)	Min-Max Fuel usage (g/ton.mile)
Flat Car	Heavy Duty Flatcar	Diesel	60,000	226	60′ 0″ (inside length)	
	89′ Flat Car	Diesel	63,000	157	–	3.2
	Center beam Flatcar 60 ft	Diesel	263,000	224,000	73′ 0″ (inside length)	
	Center beam Flatcar 73 ft.	Diesel	286,000	195,000	–	
Double stack	40′ double stack car	Diesel	–	192,000	5440	
	48′ double stack car	Diesel	–	192,000	6528	2.7–5.5
	53′ double stack car	Diesel	–	192,000	7208	

2.2.4. Waterway vessel parameters.

Table A6
Vessel and barge fuel consumption (Rodrigue, 2009; American Bureau of Shipping, 2011; Kruse et al., 2012).

Mode	Asset type	G/1000ton.mile	Max load
Water	Barge carrying with max load capacity 2000 TEU	1.74	3,000,000
Water	Container ship with max load capacity of 4000 TEU (Panamax)	0.23	104,000,000
Water	Container ship with max load capacity of 12,00 TEU (Neo Panamax)	0.19	238,000,000

2.2.5. Airplanes' parameters.

Table A7
Most common freight airplanes and their fuel consumption in the US (Airbus Company, 1993; Boeing Company, 2007)

Category	Type	Frequency in US	Actual weight (without freight)	Cargo weight/payload capacity max (lbs.)	Fuel Capacity (US gallons)	Gallon per ton.mile *	MPG	Range
Airbus	A300F4-600R	123	200,700	120,000	17,940	64.20	0.26	4657.5
Boeing	757-200SF	112	115,650	88,000	11,250	86.51	0.26	2955.5

* Derived from UPS and FedEx website and Boeing, Airbus and Douglas plane catalogs and using formulas for calculation of energy efficiency.

3. Tool Worksheets

The SySCEA tool consists of 13 sheets described below (Tables A4 and A5):

1. **Introduction:** There is a brief explanation of SySCEA and a quick user guide
2. **Input parameters:** Fuel types which are used in the supply chain, price per gallon and the average speed of each mode is entered as an initial input parameter.
3. 7. **Input – First Mile Arc, Arc 1 to Arc 5 and Last Mile Arc:** Distance, part, order, packaging and asset data for each arc (route) in the supply chain is entered here by the user. Also, the user can enter the data for each alternative arc to compare it with the current one.
   8.Results- Summary: An overview of the annual and arc by arc energy cost and transit time of the supply chain for the current situation and alternative one are displayed with different graphs.
   9.Data-Pallets: The pallet sizes and weights that are used in the supply chain and are used for calculation.
   10.Data-Containers: The container sizes and weights that are used in the supply chain and are used for calculation.
   11.Data-Cities & States: 114 cities from 48 states which are used as origin and destination of each route in the supply chain.
   12.Data-Distances: 12,329 from-to distances for all the connections between the114 cities for four modes of transportation.
   13.Data-Assets: Customized Fuel efficiency for each current arc and its alternative is calculated for each mode of transportation and is used for calculation thorough out the workbook.

An overview of worksheets in SySCEA can be seen in Figs. A2 and A3. Also, a user guide and built-in macros help the user quickly learn how to use SySCEA.



Fig. A2. SySCEA workbook: Worksheet tabs of the energy tool in Excel.

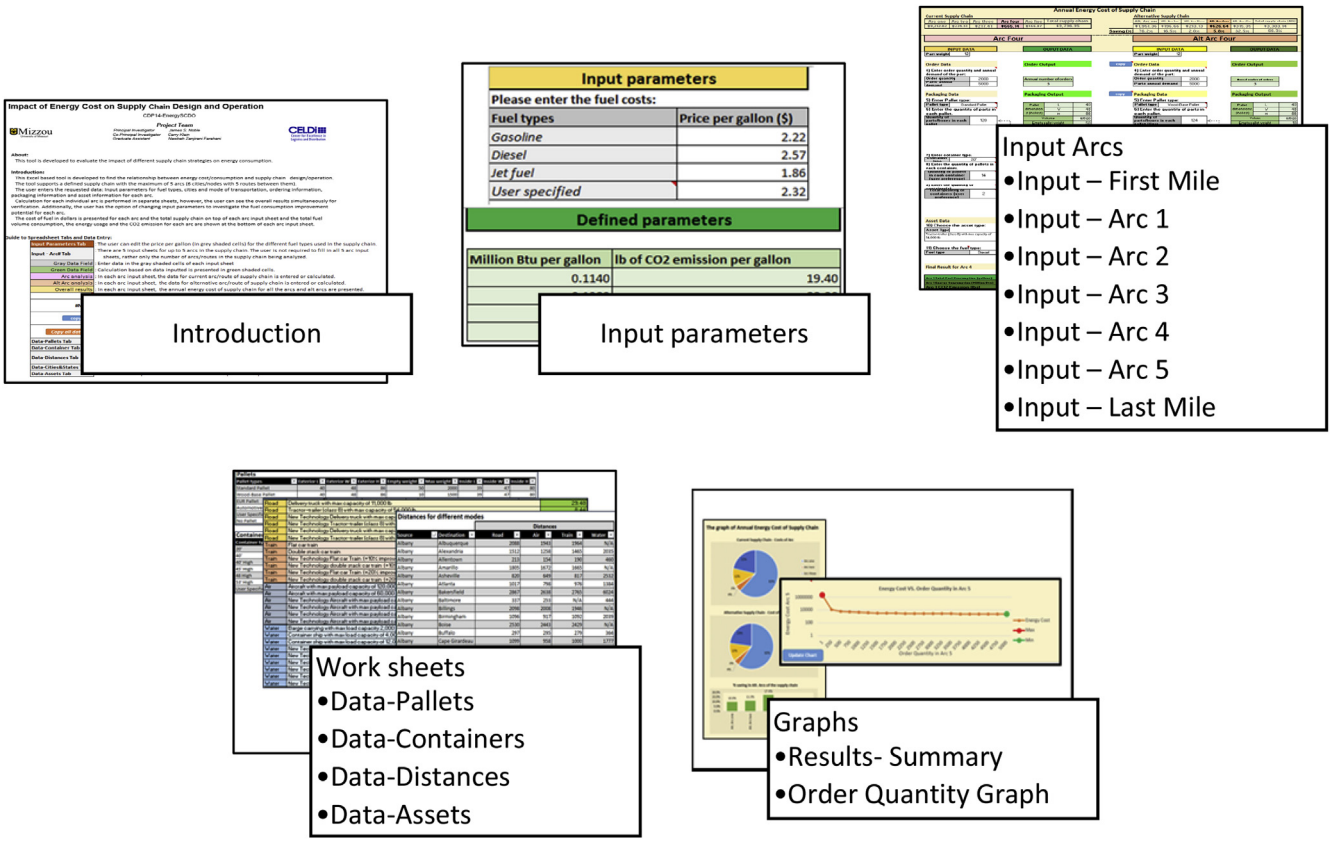


Fig. A3. An overview of the SySCEA in MS Excel.

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